POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2022
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

a) Consider the analog signal $x_a(t) = 3\cos(100\pi t)$ 8

i) Determine the minimum sampling rate required to avoid aliasing.

ii) What is the Nyquist rate for the signal $X_a(t)$?

iii) If the signal is sampled at the rate F_s = 200Hz. What is the discrete-time signal obtain after sampling?

iv) What is the frequency $0 < F < F_s/2$ of a sinusoid that yields samples identical to those obtained for F_s=200Hz

b) Define time-invariance and causality of a discrete time system. Examine whether following systems are stable or not: 7

i) $y[n] = x[n] + 1$

ii) $y[n] - y[n-1] = x[n] + x[n-1]$

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a) Find the convolution between two signals $x[n] = a^n, for\ 0 \le n \le 6$ and $h[n] = 1, for\ 0 \le n \le 4$ 8

b) Define Z transform. Find the inverse z transform of $X(z) = \log(1 + az^{-1})$ 7

a) Why do we need DFT when we have DTFT? Determine the circular convolution of the sequence 8

$$x_1(n) = \{1, 2, 3, 4\}\ and\ x_2(n) = \{5, 6, 7, 8\}$$

Also verify your answer.

b) Define Fast Fourier Transform (FFT)? Determine 8-point DFT of the sequence using Decimation in Time FFT. 7

$$x(n) = cos\left(\frac{n\Pi}{4}\right)$$

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4. a) Obtain the parallel form realization of following IIR filter

$$H(z) = \frac{1 + \frac{1}{4}z^{-1}}{(1 + \frac{1}{2}z^{-1})(1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2})}$$

b) Convert the following IIR filter into lattice ladder structure

$$H(z) = \frac{1 + z^{-1} + 2z^{-2} + z^{-3}}{1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}}$$

5. a) Convert the following in Direct form- II realization described by

$H[z] = (1+z^{-1}+2z^{-2}+3z^{-3})/(1+15z^{-1}+15z^{-2}+12z^{-3})$

b) With the low pass specification Ω_p = 3.2 KHz and Ω_s = 4.8 KHz and sampling frequency Ωfs = 12 KHz and α_s = 40 dB, find the length and value of β for Kaiser Window.

6. a) Derive Impulse Invariant Method for IIR filter design. Also illustrate the mapping from the s-plane to the z-plane while using IIM.

b) Design a Chebyshev analog filter with maximum passband attenuation of 2.5 dB at $\Omega_p = 20\ rad/sec$ and stopband attenuation of 30 dB at $\Omega_s = 50\ rad/sec$

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7. Write short notes on: (Any two)

a) Gibbs Phenomena

b) Basis Elements of DSP

c) BIBO Stable

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